Check for updates

Received: 30 November 2023 | Revised: 23 February 2024 | Accepted: 14 March 2024

DOI: 10.1111/jocd.16299

JCD Journal of Cosmetic Dermatology WILEY

REVIEW ARTICLE

Efficacy of probiotic supplementation in the treatment of psoriasis—A systematic review and meta-analysis

Kebo Wei MM[1,2] | Xiaoshu Liao BS[2] | Tianlin Yang MM[1,2] | Xin He MM[1,2] | Dongyue Yang MM[1,2] | Lingyao Lai MM[1,2] | Jing Lang MM[1,2] | Min Xiao PhD[1,2] | Jun Wang BS[2]

[1]Clinical Research on Skin Diseases, School of Clinical Medicine, Chengdu University of Traditional Chinese Medicine, Chengdu, China

[2]Department of Dermatology, Hospital of Chengdu University of Traditional Chinese Medicine, Chengdu, China

Correspondence
Min Xiao and Jun Wang, Dermatology of Department, Hospital of Chengdu University of Traditional Chinese Medicine, Chengdu 610072, China.
Email: xiaomin@cdutcm.edu.cn and 18980540226@163.com

Abstract

Objective: Many studies have emphasized the possible role of probiotics in psoriasis, probiotic supplementation might be helpful to treat psoriasis. This study systematically evaluated the efficacy of probiotic supplementation for the treatment of psoriasis.

Methods: We searched some databases with keywords until November 10, 2023, including PubMed, Cochrane Library, Embase, and Web of Science. These keywords included probiotics, psoriasis RCT, and so on. After rigorous literature screening by two authors, five studies were identified. Eventually, the required data were independently extracted by another author.

Results: A total of five studies with 286 patients were included. The pooled results showed that the efficacy of probiotic supplementation was superior to placebo in the treatment of psoriasis. The Psoriasis Area and Severity Index (SMD = −1.40, 95% CI = −2.63 to −0.17, $p < 0.00001$) and Dermatology Life Quality Index (SMD = −0.92, 95% CI = −1.86 to 0.01, $p < 0.00001$). Score decreased after probiotic supplementation.

Conclusions: The meta-analysis showed that probiotic supplementation could be a new treatment option for psoriasis.

KEYWORDS
DLQI, PASI, placebo, probiotic, randomized controlled trials

1 | INTRODUCTION

Psoriasis is an inflammatory skin disease that seriously affects the physical and mental health of humans. Some patients may also be accompanied by severe itching. It is commonly associated with genetics, immune disorders, infections, environmental factors, and endocrine factors.[1] It is reported that the global prevalence ranged from 0% to 8.5% and its prevalence is increasing in developed countries.[2–4] Research indicated that hyperactivation of some functions of the adaptive immune system is thought to be central to the pathogenesis of psoriasis.[5]

The traditional therapeutic options for psoriasis are topical corticosteroids, vitamin D analogues, calcineurin inhibitors, and keratolytics.[6–8] For moderate-to-severe plaque psoriasis, immune

Xiaoshu Liao and Kebo Wei are co-first authors and contributed equally to this work.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2024 The Authors. *Journal of Cosmetic Dermatology* published by Wiley Periodicals LLC.

modulatory agent and biologics as an option for first-line treatment.[9–11] These classic treatments are focused on reducing skin inflammation and immunomodulation. Developmental biology studies have shown that both skin cells and intestinal cells originate from the same embryonic layer and have similar signaling mechanisms. At the same time, the skin and gut are also key immune and neuroendocrine organs, and there is a close link between gut health and skin homeostasis. The "gut–skin axis" theory is a well-established theory that has been closely linked to gut health and skin homeostasis.[12] Studies have shown that gut microbiome can maintain skin immune homeostasis by mediating both innate and adaptive immunity.[12] Some studies found that oral probiotics may reverse thepro-inflammatory and anti-inflammatory immune imbalance by modulating the gut flora, thereby reducing skin inflammation of mice.[13] A study showed that probiotics can modulate immune responses by regulating the intestinal flora, thereby effectively ameliorating skin inflammation in psoriasis. Effectively improve skin inflammation in psoriasis.[14] The mechanism may be to increase the enrichment of Foxp3+ regulatory T cells to the skin and reduce apoptosis-mediated skin inflammation, thus restoring homeostasis through immunomodulation.[14] Thus, probiotics may have a therapeutic effect on psoriasis through immunomodulation of gut microbiome.

2 | MATERIALS AND METHODS

This meta-analysis was conducted based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Statement, and we have registered in the PROSPERO (CRD42023483637).

2.1 | Search strategy

We searched some databases until November 10, 2023, including Cochrane Library, Embase, Web of Science, and PubMed. Some keywords were combined to retrieve the relevant studies including medical subject heading (MeSH), title, and abstract. The following search strategy was as follows (ProbioticAND Psoriasis OR Psoriases Pustulosis of Palms).

2.2 | Eligibility criteria

2.2.1 | Inclusion criteria

Inclusion criteria included randomized controlled trials (RCTs) published in English, diagnosed with psoriasis and the test groups were treated with probiotic and the control groups were treated with placebo. The primary outcomes are Psoriasis Area and Severity Index (PASI) and Dermatology Life Quality Index (DLQI), which are the main criteria for evaluating the severity of psoriasis. The higher scores indicate more severe disease, and conversely, the lower the scores revealed the milder the disease.

2.2.2 | Exclusion criteria

Exclusion criteria including non-RCTs pregnant or lactational women and not relevant to the content of the experimental and control groups duplicated publications and animal trials and studies with unavailable data (Figure 1).

2.3 | Literature screening and data extraction

Two independent authors (LY Lai and X He) screen title and abstracts of studies from the eligibility criteria and then exclude irrelevant studies. The full text of the retained studies was then read to identify studies for inclusion. Disagreements and discrepancies were resolved by the third author. The authors (LY Lai and X He) extracted the data from included trial. All data regarding study characteristics, interventions, comparisons, and outcomes were extracted. We contacted the investigators of the studies by e-mail when necessary to obtain raw or missing data (Table 1). We also counted the probiotics included in these studies (Table 2).

2.4 | Assessment of methodological quality

We chose the Cochrane collaboration's tool[15] to evaluate the methodological quality of each included trial. The included studies were marked as low, high, or unclear risk of bias in seven items. The disagreement was resolved by discussing with the other author (Figure 2A,B).

2.5 | Statistical analysis

We chose Review Manager (RevMan) 5.3 for data analysis. Review Manager version 5.3 was used for meta-analyses. For continuous data with the same measurement unit, we used the weighted average difference and 95% confidence interval for continuous data with same measurement unit measurement units, On the contrary, we use the standardized average balance and 95% confidence interval. If $I^2 > 50\%$ or $p < 0.05$, indicating a significant heterogeneity, a random-effect model was used. Otherwise, a fixed-effect model was applied for statistical analysis.[16,17] When 10 or more studies were included in the same outcome, the potential publication bias was assessed by using funnel plot. A two-tailed $p < 0.05$ was considered to be statistically significant.

3 | RESULTS

We retrieved 454 studies from four literature databases. After removing duplicated literatures, 78 studies remained for further selection. Thirty-eight studies were removed by screening their abstract and title. Thirty-five studies were excluded by reading full text.

14732165, 2024, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jocd.16299, Wiley Online Library on [14/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

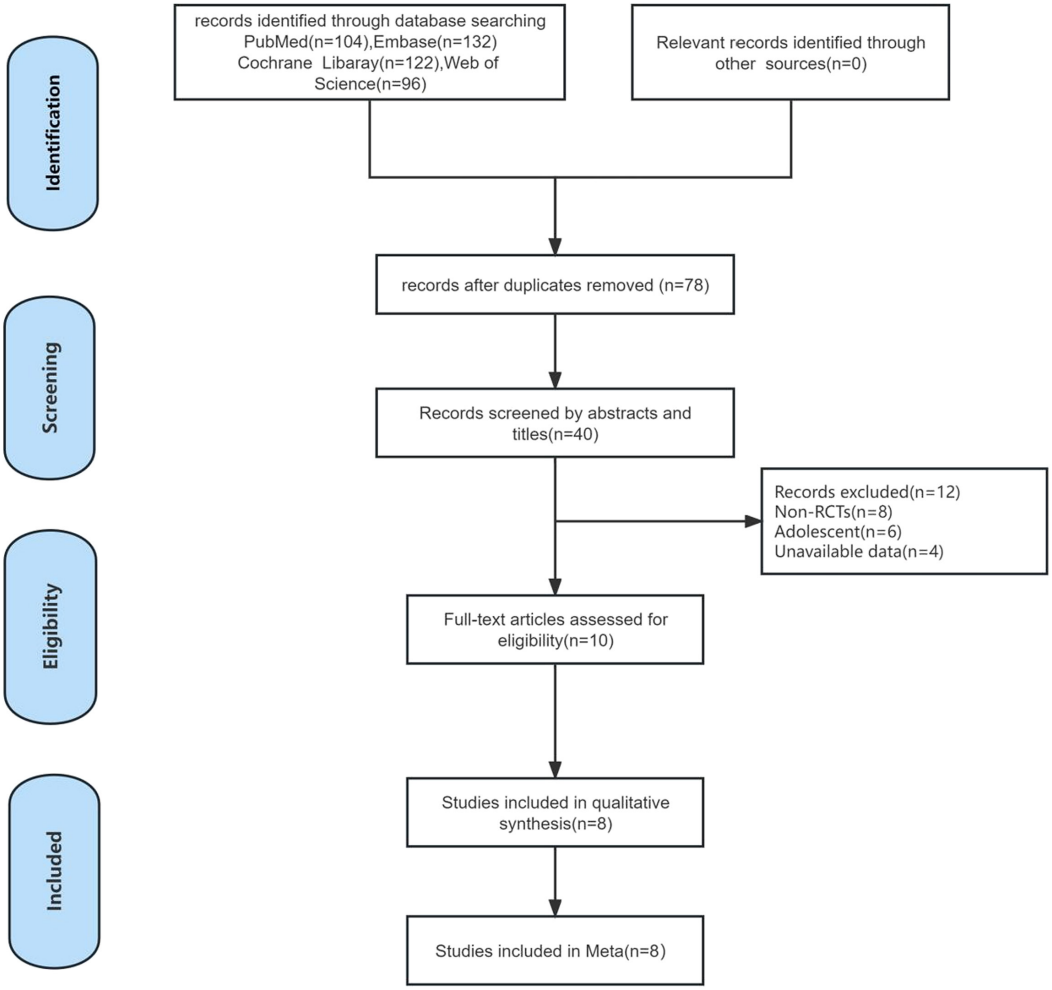


FIGURE 1 Selection of the included studies.

TABLE 1 Characteristics of the included studies.

Study id	Sample size		Age (years)	Interventions		Duration (months)	Outcomes
	T	C		T	C		
Enrico Stefan Suriano et al. 2022	50	53	18	Standard-of-care plus probiotics	Standard-of-care plus placebo	6	PASI DLQI
Jalal Moludi et al. 2021	25	25	18	Probiotic capsule	Maltodextrin capsules	2	PASI DLQI
Ali Akbarzadeh et al. 2022	27	25	18	Lactocare® hydrocortisone	Hydrocortisone placebo	3	PASI DLQI
Isadora O. Gilli, MD et al. 2023	18	17	18	Probiotic	Placebo	2	PASI DLQI
Jalal Moludi et al. 2022	23	23	18	Probiotic	Placebo	2	PASI

Abbreviations: C, control group; DLQI, Dermatology Life Quality Index; PASI, Psoriasis Area and Severity Index; T, test group.

Ultimately, five studies were included in this study. The flowchart of the study selection process is shown in Figure 1. Figure 2A shows the risk of bias of randomized controlled trials, because of all studies did not report adverse events and elective reporting were marked as unclear risk, in terms of other bias, all studies did were rated as low risk. Table 1 show the characteristics of the included studies. Table 2 show the types of probiotics of the included studies, which contain *Lactobacillus rhamnosus*, *Bifidobacterium*, *Lactobacillus acidophilus*, and *Streptococcus thermophilus*.

14732165, 2024, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jocd.16299, Wiley Online Library on [14/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 The probiotics of the included studies.

Study id	Probiotics
Enrico Stefan Suriano et al. 2022	*Lactobacillus rhamnosus* formula
Jalal Moludi et al. 2021	Probiotic capsule (*Lactobacillus acidophilus*, *Bifidobacterium bifidum*, *Bifidobacterium lactis* and *Bifidobacterium langum*)
Ali Akbarzadeh et al. 2022	Lactocare® (*Lactobacillus* strains, Bifido-bacteria strains, and *Streptococcus thermophilus*, plus Fructo-oligosaccharides)
Jalal Moludi et al. 2022	Probiotic capsule (*L. acidophilus*, *B. bifidum*, *B. lactis* and *B. langum*)
Isadora O. Gilli, MD et al. 2023	*Lactobacillus rhamnosus*

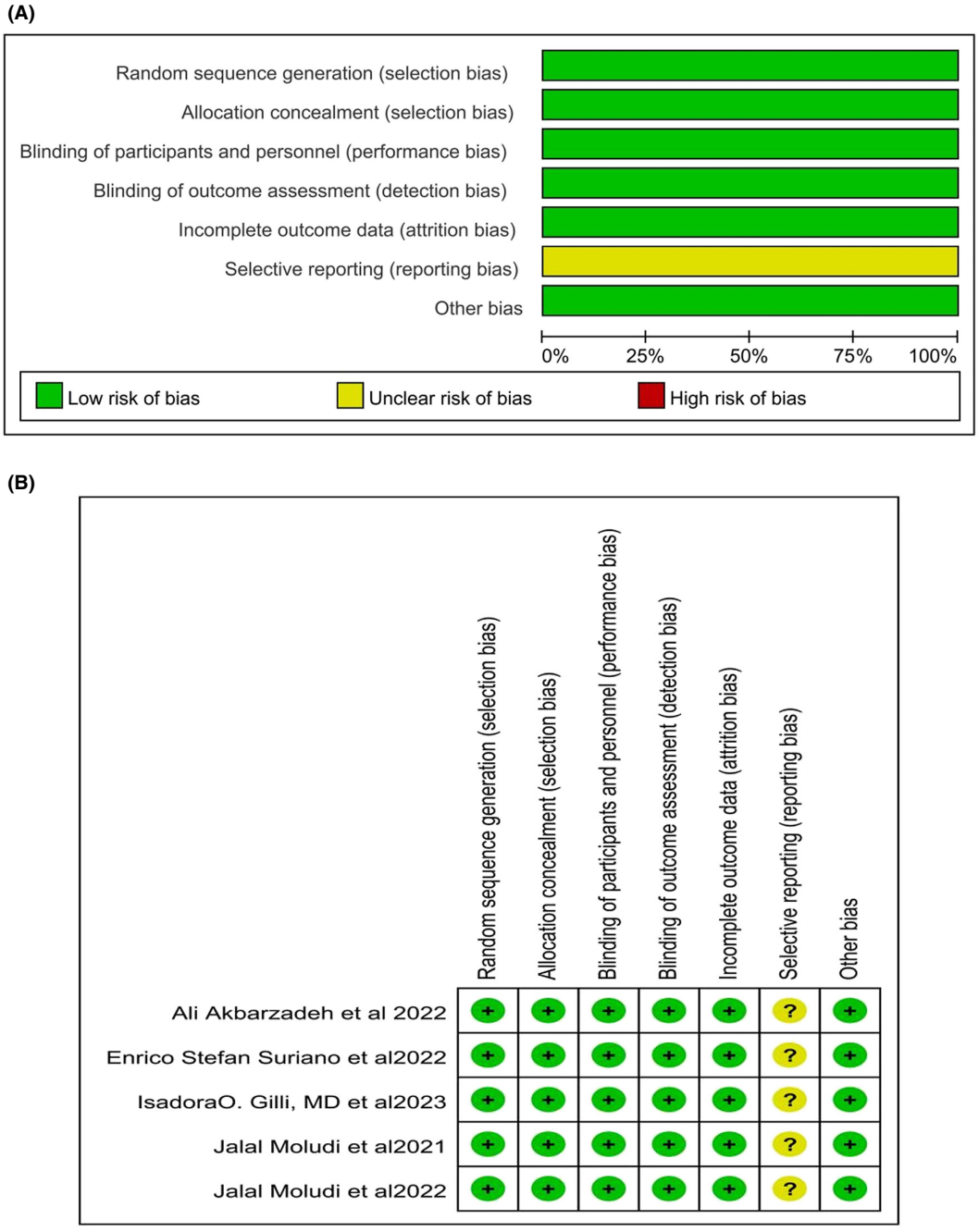


FIGURE 2 Cochrane risk of bias assessment of RCT.

14732165, 2024, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jocd.16299, Wiley Online Library on [14/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3.1 | Outcomes

3.1.1 | Psoriasis Area and Severity Index

All studies[18–22] were reported the PASI scores, because the heterogeneity among the studies was not significant ($p = 0.001$, $I^2 = 78\%$), we chose a random-effect model. The results showed that PASI scores were significantly lower in the probiotic supplement group (SMD = −1.40, 95% CI = −2.63 to −0.17, $p < 0.00001$) (Figure 3).

3.1.2 | Dermatology Life Quality Index

All studies[18–22] were reported the DLQI scores, because the heterogeneity among the studies was not significant ($p = 0.09$, $I^2 = 50\%$), a fixed-effect model was performed. DLQI scores also showed lower after probiotic supplement group (SMD = −0.92, 95% CI = −1.86 to 0.01, $p < 0.00001$) (Figure 4).

3.1.3 | Publication bias

All outcomes were constrained by the number of studies ($n < 10$), so we did not adopt publication bias detection.

4 | DISCUSSION

In this study, we reviewed and comprehensively summarized randomized controlled, indicating some important information. This meta-analysis revealed that PASI and DLQI score were decreased after probiotic supplementation as a treatment for psoriasis. Psoriasis is a common immune-mediated chronic inflammatory skin disease that can be triggered by genetic and immunologic, infected, and metabolic disorders.[23]

The association between probiotic deficiency and psoriasis uncertain, but some studies have suggested possible roles of probiotic in psoriasis. Some studies demonstrated that gut microbiota is associated with inflammatory dermatosis.

Microbiome dysbiosis may lead increased epithelial permeability of the intestinal mucosa and activation of effector T cells disturb the balance between them and regulatory T cells, while proinflammatory cytokines further enhance epithelial permeability and create chronic systemic inflammation.[24] Some studies reported that two beneficial bacterial species (*Parabacteroides* and *Coprobacillu*) is obviously decreased in the patients of psoriatic arthritis.[25] An animal experiment indicated Oral supplementation of psoriasis-like mice with probiotics significantly modified their psoriasis-like pathological phenotype and inhibited the release of inflammatory cytokines associated with the IL-23 /Th17 cell axis.[26] These research results showed gut microbiota play an important

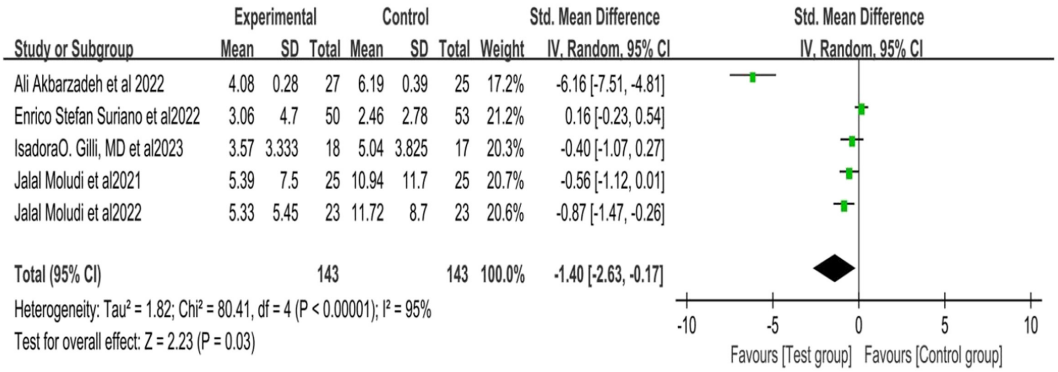


FIGURE 3 Psoriasis Area and Severity Index (PASI).

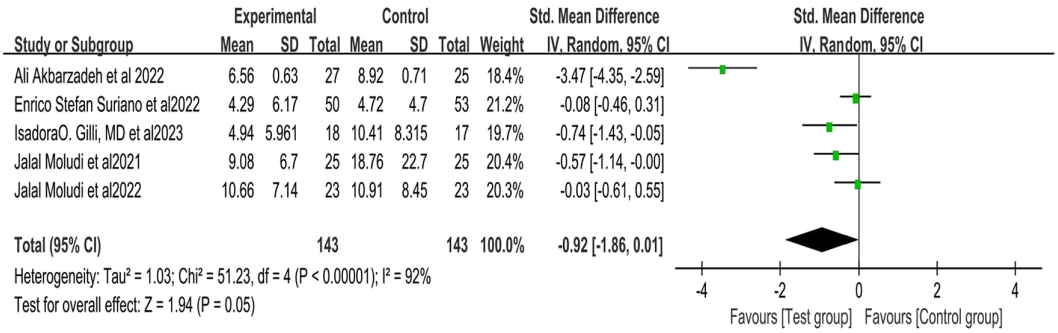


FIGURE 4 Dermatology Life Quality Index (DLQI).

14732165, 2024, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jocd.16299, Wiley Online Library on [14/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

role in psoriasis. Besides, a series of clinical researches proved that probiotic supplement could apparently alleviate symptom and decrease in skin inflammatory markers, such as TNF-α, IL-6, and IL-17 of psoriasis.[27–29] The mechanism of action of probiotic supplementation in patients with psoriasis may be related to increased levels of acetate And propionate in the gut, which have been linked to the suppression of cytokine activity.[30]

The study included five randomized double-blind clinical trials and concluded that probiotic supplementation decreased the PASI and DLQI score and probiotic supplementation benefits psoriasis patients. Meanwhile, more and more researches showed that psoriasis is always associated with metabolic syndrome. Psoriasis also is considered as an independent risk factor for cardiovascular disease.[31] Some researches found probiotic supplements, such as inulin and fructooligosaccharides could decrease cholesterol and triglyceride levels in both in vivo and in vitro studies.[32] A study also reported probiotic supplements decreased uric acid.[32] In conclusion, probiotic supplementation could not only alleviate the symptom of psoriasis but also decrease cholesterol and triglyceride levels and uric acid. Several studies have shown a relationship between gut flora and skin.[33] Moreover, dysregulation of microbiota has been seen in some inflammatory skin conditions, such as psoriasis, acne, seborrheic dermatitis, and atopic dermatitis.[34] A number of researches have showed that probiotic supplements is helpful in treatment of inflammatory skin conditions, including atopic dermatitis, acne, seborrheic dermatitis, and psoriasis.[35–39] The experimental group in this study contained a variety of probiotics, including single-strain probiotics and multi-strain mixtures probiotics.

These probiotics have shown good efficacy in the treatment of psoriasis and some research showed that in most cases, multi-strain mixtures were not significantly more effective than single-strain probiotics.[40] Although, biologics have become the mainstay of treatment for psoriasis, but the disease requires long-term maintenance therapy, and probiotics have some efficacy as a complementary treatment and appear to be better at stabilizing the disease in maintenance therapy. Of course, more clinical studies are needed to confirm these conclusions.

However, we also have some limitations in this study.

First, the number of included studies and sample were relatively small, the results need to prove by lots of multi-center, large-sample clinical studies. Second, the efficacy of the meta is only based on clinical scores, we need more laboratory indicators to demonstrate it.

5 | CONCLUSION

We systematically and comprehensively reviewed the relevant studies, these studies revealed probiotic is helpful in the treatment of psoriasis and the results of this meta summarized the evidence for the role of probiotic in psoriasis patients in comparison with a placebo group, probiotic supplementation decreased psoriasis severity and improved the symptoms and clinical signs of psoriasis. However, the specific mechanisms underlying are still unclear. More high-quality, large-sample, and long-term trials are required to figure out the underlying mechanisms and provide more reliable and accurate efficacy evidence in the future.

AUTHOR CONTRIBUTIONS

KW designed the study, created the tables and pictures, and wrote the manuscript. TY, XH, and DY conducted the literature searches, data extraction, methodological quality evaluation, and statistical analysis. JL, LL and MX helped to revise the manuscript. All authors approved the final version of the manuscript.

CONFLICT OF INTEREST STATEMENT

No potential conflict of interest was reported by the author(s).

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this published article.

ORCID

Kebo Wei https://orcid.org/0009-0006-8427-8166

REFERENCES

1. Armstrong AW, Read C. Pathophysiology, clinical presentation, and treatment of psoriasis: a review. *JAMA*. 2020;323(19):1945-1960.
2. Parisi R, Symmons DP, Griffiths CE, Ashcroft DM, Identification and Management of Psoriasis and Associated ComorbidiTy (IMPACT) Project Team. Global epidemiology of psoriasis: a systematic review of incidence and prevalence. *J Invest Dermatol*. 2013;133(2):377-385.
3. World Health Organization. *Global Report on Psoriasis*. World Health Organization; 2016. Accessed February 13, 2020. https://www.who.int/publications/i/item/9789241565189
4. Rachakonda TD, Schupp CW, Armstrong AW. Psoriasis prevalence among adults in the United States. *J Am Acad Dermatol*. 2014;70(3):512-516.
5. Nestle FO, Kaplan DH, Barker J. Mechanisms of disease: psoriasis. *N Engl J Med*. 2009;361(5):496-509.
6. Lebwohl M, Menter A, Weiss J, et al. Calcitriol 3 microg/g ointment in the management of mild to moderate plaque type psoriasis: results from 2 placebo-controlled, multicenter, randomized double-blind, clinical studies. *J Drugs Dermatol*. 2007;6(4):428-435.
7. Soleymani T, Hung T, Soung J. The role of vitamin D in psoriasis: a review. *Int J Dermatol*. 2015;54(4):383-392.
8. Gribetz C, Ling M, Lebwohl M, et al. Pimecrolimus cream 1% in the treatment of intertriginous psoriasis: a double-blind, randomized study. *J Am Acad Dermatol*. 2004;51(5):731-738.
9. Kimball AB, Rothman KJ, Kricorian G, et al. OBSERVE-5: observational postmarketing safety surveillance registry of etanercept for the treatment of psoriasis final 5-year results. *J Am Acad Dermatol*. 2015;72(1):115-122.
10. Menter A, Thaçi D, Wu JJ, et al. Long-term safety and effectiveness of adalimumab for moderate to severe psoriasis: results from 7-year interim analysis of the ESPRIT registry. *Dermatol Ther (Heidelb)*. 2017;7(3):365-381.
11. Mastorino L, Dapavo P, Susca S, et al. Drug survival and clinical effectiveness of secukinumab, ixekizumab, brodalumab, guselkumab, risankizumab, tildrakizumab for psoriasis treatment. *J Dtsch Dermatol Ges*. 2023;22:34-42. doi:10.1111/ddg.15251

14732165, 2024, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jocd.16299, Wiley Online Library on [14/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

12. O'Neill CA, Monteleone G, McLaughlin JT, Paus R. The gut-skin axis in health and disease: a paradigm with therapeutic implications. *Bioessays*. 2016;38(11):1167-1176.
13. Holowacz S, Blondeau C, Guinobert I, Guilbot A, Hidalgo S, Bisson JF. *Lactobacillus salivarius* LA307 and *Lactobacillus rhamnosus* LA305 attenuate skin inflammation in mice. *Benef Microbes*. 2018;9(2):299-309.
14. Atabati H, Esmaeili SA, Saburi E, et al. Probiotics with ameliorating effects on the severity of skin inflammation in psoriasis: evidence from experimental and clinical studies. *J Cell Physiol*. 2020;235(12):8925-8937.
15. Higgins JP, Altman DG, Gøtzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ*. 2011;343:d5928.
16. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ*. 2003;327:557-560.
17. Egger M, Smith G, Altman DG. *Systematic Reviews in Health Care: Meta-Analysis in Context*. 2nd ed. BMJ Publishing Group; 2001.
18. Suriano ES, Souza MDM, Kobata CM, Santos FHY, Mimica MJ. Efficacy of an adjuvant *Lactobacillus rhamnosus* formula in improving skin lesions as assessed by PASI in patients with plaque psoriasis from a university-affiliated, tertiary-referral hospital in São Paulo (Brazil): a parallel, double-blind, randomized clinical trial. *Arch Dermatol Res*. 2023;315(6):1621-1629.
19. Moludi J, Khedmatgozar H, Saiedi S, Razmi H, Alizadeh M, Ebrahimi B. Probiotic supplementation improves clinical outcomes and quality of life indicators in patients with plaque psoriasis: a randomized double-blind clinical trial. *Clin Nutr ESPEN*. 2021;46:33-39.
20. Akbarzadeh A, Alirezaei P, Doosti-Irani A, Mehrpooya M, Nouri F. The efficacy of Lactocare® synbiotic on the clinical symptoms in patients with psoriasis: a randomized, double-blind, placebo-controlled clinical trial. *Dermatol Res Pract*. 2022;2022:4549134.
21. Kageyama Y, Shimokawa Y, Kawauchi K, et al. Dysbiosis of oral microbiota associated with palmoplantar pustulosis. *Dermatology*. 2021;237(3):347-356.
22. Moludi J, Fathollahi P, Khedmatgozar H, et al. Probiotics supplementation improves quality of life, clinical symptoms, and inflammatory status in patients with psoriasis. *J Drugs Dermatol*. 2022;21(6):637-644.
23. Raharja A, Mahil SK, Barker JN. Psoriasis: a brief overview. *Clin Med (Lond)*. 2021;21(3):170-173.
24. Brown EM, Kenny DJ, Xavier RJ. Gut microbiota regulation of T cells during inflammation and autoimmunity. *Annu Rev Immunol*. 2019;37:599-624.
25. Scher JU, Ubeda C, Artacho A, et al. Decreased bacterial diversity characterizes the altered gut microbiota in patients with psoriatic arthritis, resembling dysbiosis in inflammatory bowel disease. *Arthritis Rheumatol*. 2015;67(1):128-139.
26. Lu W, Deng Y, Fang Z, et al. Potential role of probiotics in ameliorating psoriasis by modulating gut microbiota in imiquimod-induced psoriasis-like mice. *Nutrients*. 2021;13(6):2010.
27. Navarro-López V, Martínez-Andrés A, Ramírez-Boscá A, et al. Efficacy and safety of oral administration of a mixture of probiotic strains in patients with psoriasis: a randomized controlled clinical trial. *Acta Derm Venereol*. 2019;99(12):1078-1084.
28. Buhaș MC, Candrea R, Gavrilaș LI, et al. Transforming psoriasis care: probiotics and prebiotics as novel therapeutic approaches. *Int J Mol Sci*. 2023;24(13):11225.
29. Lin C, Zeng T, Deng Y, Yang W, Xiong J. Treatment of psoriasis vulgaris using *Bacteroides fragilis* BF839: a single-arm, open preliminary clinical study. *Sheng Wu Gong Cheng Xue Bao*. 2021;37(11):3828-3835.
30. Dei-Cas I, Giliberto F, Luce L, Dopazo H, Penas-Steinhardt A. Metagenomic analysis of gut microbiota in non-treated plaque psoriasis patients stratified by disease severity: development of a new Psoriasis-Microbiome Index. *Sci Rep*. 2020;10(1):12754.
31. Palmer V, Cornier MA, Waring A, Valdebran M. Evaluation and treatment of metabolic syndrome and cardiovascular disease in adult patients with psoriasis. *Int J Dermatol*. 2023;62(12):1437-1446.
32. Ooi LG, Liong MT. Cholesterol-lowering effects of probiotics and prebiotics: a review of in vivo and in vitro findings. *Int J Mol Sci*. 2010;11:2499-2522.
33. Thye AY, Bah YR, Law JW, et al. Gut-skin axis: unravelling the connection between the gut microbiome and psoriasis. *Biomedicine*. 2022;10(5):1037.
34. De Pessemier B, Grine L, Debaere M, Maes A, Paetzold B, Callewaert C. Gut-skin axis: current knowledge of the interrelationship between microbial dysbiosis and skin conditions. *Microorganisms*. 2021;9(2):353.
35. Husein-ElAhmed H, Steinhoff M. Effects of probiotic supplementation in adult with atopic dermatitis: a systematic review with meta-analysis. *Clin Exp Dermatol*. 2023;14:46-52.
36. Xue X, Yang X, Shi X, Deng Z. Efficacy of probiotics in pediatric atopic dermatitis: a systematic review and meta-analysis. *Clin Transl Allergy*. 2023;13(7):e12283.
37. Sánchez-Pellicer P, Navarro-Moratalla L, Núñez-Delegido E, Ruzafa-Costas B, Agüera-Santos J, Navarro-López V. Acne, microbiome, and probiotics: the gut-skin axis. *Microorganisms*. 2022;10(7):1303.
38. Polak K, Jobbágy A, Muszyński T, et al. Microbiome modulation as a therapeutic approach in chronic skin diseases. *Biomedicine*. 2021;9(10):1436.
39. Choy CT, Chan UK, Siu PLK, et al. A novel E3 probiotics formula restored gut dysbiosis and remodelled gut microbial network and Microbiome Dysbiosis Index (MDI) in southern Chinese adult psoriasis patients. *Int J Mol Sci*. 2023;24(7):6571. doi:10.3390/ijms24076571
40. Medina M, De Palma G, Ribes-Koninckx C, Calabuig M, Sanz Y. *Bifidobacterium* strains suppress in vitro the pro-inflammatory milieu triggered by the large intestinal microbiota of coeliac patients. *J Inflamm (Lond)*. 2008;5:19. doi:10.1186/1476-9255-5-19

How to cite this article: Wei K, Liao X, Yang T, et al. Efficacy of probiotic supplementation in the treatment of psoriasis—A systematic review and meta-analysis. *J Cosmet Dermatol*. 2024;23:2361-2367. doi:10.1111/jocd.16299

14732165, 2024, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jocd.16299, Wiley Online Library on [14/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License